

Magic Metrics: The Key to Harry Potter's Success

Harry Potter is a widely known children's book series (children age 9 - 12) by the author J.K Rollings (Barnes and Noble) published between 1998 and 2007. With the literary success the series achieved, it was soon adapted into 8 movies released from 2001 through 2011. The movie adaptation of the series went on to have incredible success by reaching \$1 billion dollars in the box office twice (Guerrasio, 2024); with 'Harry Potter and the Philosopher's Stone' closing with approximately \$1,002,000,000 and 'Harry Potter and the Deathly Hallows Part 2' closing with approximately \$1,342,000,000 world wide.

Given its success, this analysis aims to explore the factors contributing to Harry Potter's cinematic achievements. Three datasets were gathered to explore different metrics that may or may not have a correlation with the blockbuster's cinematic accomplishment. The objective of this analysis was to look into the relationship between the number of unique characters and box office performance, and the global IMDb ratings of the franchise. It was hypothesized that there will not be a significant relationship between the number of unique characters and the franchise's success; measured by box office revenue, global audience ratings, and budget.

DATASET OVERVIEW

The analysis utilized the 'hp_merged' dataset which was composed of the 'Harry Potter Movies' dataset (sourced from Kaggle) ,and two IMDb Non-Commercial Datasets: 'akas' and 'ratings.' The initial datasets included information on characters, dialogue, chapters, movies, production budgets, box office revenue, alternative movie titles, regions, languages, global audience ratings, and vote counts.

To prepare the data, columns were standardized, merged based on shared keys, and filtered to include only relevant rows. Special characters (e.g., '\$' and ',') were removed from numerical columns such as budget and box office revenue, and data types were converted for analysis. After removing unnecessary columns like 'Runtime,' 'House,' and 'Patronus,' the final dataset contained 11 columns and 6,432 rows. Key features of the dataset included box office revenue, production budgets, IMDb global ratings, and the number of unique characters per movie.

EXPLORATORY DATA ANALYSIS (EDA)

Before diving into the merged dataset, the number of unique characters were counted and the 'Character ID' column was renamed 'Number of Unique Characters' and placed back into 'hp_merged'. The merged dataset columns were rechecked and repeating columns were dropped.

The first step in the EDA was to look at the summary statistics to obtain a deeper understanding of the data's key variables and the distributions. This was done using histograms, which provided insight into the data's distribution and any potential outliers. The histogram of the box office revenue showed that the majority of the box office revenue was concentrated in the lower range clustered below \$1 billion with the exception of one outlier. This was indicative of a skewed distribution of the box office revenue. Similarly, the budget histogram also displayed a skewed distribution, with a concentration of movies in the lower values and a few movies with a higher budget. The histogram for average rating was evenly spread with the exception of a notable decrease in audience ratings towards the movies later on in the series. The majority of the films fell between a rating of 6.6 and 7.7. The histogram for the number of unique characters indicates that the number of characters per movie stay mostly consistent with the exception of a

few films slightly increasing; likely due to the story expanding. To further investigate the relationship between the key variables, a pair plot was generated to obtain a comprehensive view of the relationships between each pair in the dataset.

To confirm the presence of outliers in the data, a z-score was used to check every column in 'hp_merged' for outliers. For each variable, we calculated the z-score and flagged any values with a z-score greater than 2.5 or less than -2.5, indicating potential outliers. The movie "Harry Potter and the Deathly Hallows Part 2" was identified as an outlier due to its exceptionally high box office revenue combined with a relatively low IMDb rating. Since this outlier significantly contributed to the overall analysis, it was kept in the dataset for further exploration rather than discarded.

STATISTICAL ANALYSIS

Pearson's correlation coefficient was used to explore the relationships between key variables, visualized through a heat map. The results revealed notable correlations: a strong negative correlation between box office revenue and average global ratings (-0.95), indicating that higher revenue is associated with lower IMDb ratings; a moderate positive correlation between budget and box office revenue (0.59), suggesting that bigger budgets tend to yield higher financial success; and a weak positive correlation between the number of unique characters and box office revenue (0.21), indicating minimal influence. Weak negative correlations were observed between audience ratings and both the number of unique characters (-0.28) and budget (-0.40), as well as between budget and the number of unique characters (-0.30). These findings suggest that box office revenue is not a reliable indicator of audience ratings, and higher budgets improve financial success without necessarily increasing ratings or

character diversity. Overall, the franchise's cinematic success likely stemmed from factors beyond the story, reflecting a complex relationship between financial investment, marketing, and audience reception.

In the final step of the statistical analysis, the hypothesis was tested using p-values to look deeper into the importance of the relationships between the variables examined. The p-values demonstrated statistically significant results across all comparisons. For instance, the p-value for the correlation between Box Office and averageRating is 0.0, indicating an extremely strong statistical significance, meaning there is a clear relationship between these variables. Similarly, the p-value between Box Office and Budget is also 0.0, emphasizing the significance of budget in driving box office success. The relationship between Box Office and Number of Unique Characters, though weaker, remains highly significant with a p-value of approximately $3.04e-66$. Additionally, the p-values for the relationships between Number of Unique Characters and other variables (e.g., averageRating and Budget) are extremely small (e.g., $3.19e-116$ and $1.89e-137$), highlighting strong statistical evidence for the presence of relationships, even if the correlations themselves are weak. Across all comparisons, the low p-values (<0.05 or effectively 0) confirm that the observed relationships are unlikely to be due to random chance, supporting the robustness of the analysis. These results suggest that each variable, to varying degrees, contributes to the dynamics of Harry Potter's cinematic and audience reception metrics. Therefore, rejecting the null hypothesis and accepting the alternative hypothesis.

CONCLUSION

The statistical analysis conducted on the 'hp-merged' dataset explored the relationships between the number of unique characters, box office performance, global IMDb ratings, and

production budgets within the Harry Potter franchise. The results revealed statistically significant relationships between these variables, though the strength and direction of the correlations varied. Future research could benefit from applying a linear regression model to better understand the data used and how these variables generalize throughout the film industry. Additionally, comparing the Harry Potter franchise to other successful franchises—particularly those adapted from literature—could provide deeper insights into the factors that contribute to both commercial and critical success.

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